Year 13 Further Maths - Series

Similar questions - Using standard results for sum r, sum r^2

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Using the standard formulae for $\sum r$ and $\sum r^2$, show that $\sum_{r=1}^{12} r(2r+1) = 1378$. **[3]**

(b) Show that $\sum_{r=1}^{n} r(r+2) = \frac{n(n+1)(2n+7)}{6}$. **[4]**

2

(a) In this part you must show detailed reasoning. Determine the value of $\sum_{r=1}^{80} (3r-1)^2$. **[3]**

(b) The series S is the sum of the squares of all positive odd integers from 1^2 to 59^2. Determine the value of S. **[5]**

(c) Determine the least value of n for which $\frac{1^2+2^2+\ldots+n^2}{1+2+\ldots+n} > 20$. **[4]**

Question 1

(a) $\sum r(2r+1) = 2\sum r^2 + \sum r = 2 \times \frac{12 \cdot 13 \cdot 25}{6} + \frac{12 \cdot 13}{2} = 2 \times 650 + 78 = 1378$

(b) $\sum r(r+2) = \frac{n(n+1)(2n+1)}{6} + n(n+1) = \frac{n(n+1)}{6}[(2n+1)+6] = \frac{n(n+1)(2n+7)}{6}$

Question 2

(a) $\sum(9r^2 - 6r + 1) = 9 \cdot \frac{80 \cdot 81 \cdot 161}{6} - 6 \cdot \frac{80 \cdot 81}{2} + 80 = 1564920 - 19440 + 80 = 1545560$

(b) The odd integers are $2r-1$ for $r = 1, \ldots, 30$ (since $2 \times 30 - 1 = 59$). $\sum(2r-1)^2 = \sum(4r^2 - 4r + 1) = 4 \cdot \frac{30 \cdot 31 \cdot 61}{6} - 4 \cdot \frac{30 \cdot 31}{2} + 30 = 37820 - 1860 + 30 = 35990$

(c) $\dfrac{n(n+1)(2n+1)/6}{n(n+1)/2} = \dfrac{2n+1}{3} > 20$, so $n > 29.5$ and the least value is $n = 30$